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Profile

I work on data-driven and model-based control strategies such as reinforcement learning (RL) and model predictive control (MPC), along with machine learning methods, to make buildings, energy systems, and other AI-driven applications smarter and more efficient. I bring experience from both research and industry to solving practical engineering challenges.

Education

- Purdue University, IN, United States** 2023 –
PhD in Mechanical Engineering — **GPA:** 3.84/4.0
Minor in Computational Science and Engineering
Advisor: [Kevin J. Kircher](#)
- University of British Columbia, BC, Canada** 2021 – 2023
MS in Mechanical Engineering — **GPA:** 4.33/4.33 (94%)
Thesis: [Mixing gaseous hydrogen into natural gas distribution pipelines](#)
Advisor: [Sunny R. Li](#)
- University of Tabriz, EA, Iran** 2016 – 2020
BS in Mechanical Engineering — **GPA:** 4.0/4.0 (19.12/20) — **Rank:** 1/155+

Publications — Full list on [Google Scholar](#)

Journal articles

2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, P. Papageorgiou, A.H. Lee, J.E. Braun, G.P. Henze, and K.J. Kircher, “Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review,” *Applied Energy*, 2025. [\[DOI\]](#)
1. **A.J. Khabbazi**, M. Zabihi, R. Li, M. Hill, V. Chou, and J. Quinn, “Mixing hydrogen into natural gas distribution pipeline system through Tee junctions,” *International Journal of Hydrogen Energy*, 2024. [\[DOI\]](#)

Conference papers

6. **A.J. Khabbazi** and K.J. Kircher, “Small HVAC control demonstrations in larger buildings often overestimate savings,” *under review*.

5. E.N. Pergantis, L.D. Reyes Premer, **A.J. Khabbazi**, Priyadarshan, F. Wu, D. Ziviani, and K.J. Kircher, “Active current limiting control of residential appliances for breaker panel protection across the US: a parametric study,” *CISBAT*, 2025, pp. 1–6.
4. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, A.H. Lee, J. Ma, H. Liu, G.P. Henze, K.J. Kircher, “What Have We Learned From Field Demonstrations of Advanced Commercial HVAC Control?,” *International High Performance Buildings Conference*, 2024, pp. 1–10. [\[DOI\]](#)
3. **A.J. Khabbazi**, M. Zabihi, R. Li, V. Chou, and J. Quinn, “Blending of Hydrogen into a Natural Gas Distribution Pipeline in British Columbia through a Tee Junction for Reducing GHG Emissions,” *Canadian Society for Mechanical Engineering International Congress*, 2023, pp. 1–6. [\[DOI\]](#)
2. **A. Khabbazi**, R. Li, and J. Quinn, “Green Hydrogen Supply to Urban Infrastructure and Buildings through Blending into the Existing Grid,” *Canadian Society for Mechanical Engineering International Congress*, 2022, pp. 1. [\[DOI\]](#)
1. **A. Khabbazi**, R. Li, and J. Quinn, “The Blending and Transmission of Hydrogen and Natural Gas in Transmission and Distribution Pipelines,” *International Green Energy Conference (IGEC-XIII)*, 2021, pp. 1. [\[DOI\]](#)

Industry experience

Summer Intern, Tesla, Reno, NV, United States 05/2025 – 08/2025

- Built supervised ML models (XGBoost, neural nets, SVR; ensembles) for defect detection and quality assessment.
- Supported deployment of a thermal imaging system for non-destructive inspection in drive-unit manufacturing.

- Built data pipelines with MySQL and applied feature engineering to identify high-signal inputs for ML development.
- Contributed to projects projected to reduce testing costs by over \$1M annually through improved quality assessment.

Student Researcher, [FortisBC](#), *Vancouver, BC, Canada*

09/2021 – 09/2023

- Supported assessments on hydrogen injection into natural gas pipelines and renewable gas projects (UBC H₂Lab).
- Collaborated with engineers and research teams (materials, sensors, AI, safety) to assess industry impacts.

Research experience

PhD Research Assistant, Purdue University

09/2023 –

- Supported commissioning of Herrick Labs by upgrading the BAS (Tridium Niagara) and integrating Modbus systems.
- Developed ML models for occupancy detection (98% accuracy) and real-time energy/comfort dashboards (InfluxDB).
- Designed HVAC control strategies with MPC and DeePC, showing DeePC matched MPC without a system model.
- Reviewed 100+ publications on MPC and RL in HVAC, leading to a published [review paper](#) in Applied Energy.

MS Research Assistant, University of British Columbia

07/2021 – 09/2023

- Researched hydrogen injection into natural gas pipelines using CFD, real-gas modeling (C), and CAD design.
- Analyzed 1 TB of simulation data, leading to a [journal paper](#), three conference papers, and two awards.

Skills

Programming: Python, C, MATLAB, R, SQL (MySQL), Git, Linux

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, Keras

Data & Analytics: NumPy, Pandas, SciPy, Matplotlib, Seaborn, Grafana

Optimization & Modeling: CVX, CVXPY

Systems: Tridium Niagara (BAS), InfluxDB (TSDB), Modbus, IO-Link, MTConnect

Selected Honors & Awards

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| • ASHRAE Graduate Student Grant-in-Aid Award | 2025 |
| • Best Paper Award at CSME 2023 Conference | 2023 |
| • Best Presentation Award at CSME 2022 Conference, Advanced Energy Symposium | 2022 |
| • UBC Graduate Scholarship | 2022 |
| • UBC Dean's Entrance Scholarship | 2021 |

Conferences

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| 8th International High Performance Buildings Conference , <i>West Lafayette, USA</i> | 07/2024 |
| ASHRAE Winter Conference , <i>Chicago, USA</i> | 01/2024 |
| Canadian Society for Mechanical Engineering 2023 Conference , <i>Sherbrooke, Canada</i> | 05/2023 |
| Canadian Society for Mechanical Engineering 2022 Conference , <i>Edmonton, Canada</i> | 07/2022 |
| 13th International Green Energy Conference , <i>Virtual</i> | 07/2021 |

Workshops & seminars

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| Intelligent Building Operations (IBO) Workshop , <i>West Lafayette, USA</i> | 07/2024 |
| What have we learned from field demonstrations of advanced commercial HVAC control? [Recording] | |

Selected Courses

- Optimization, Control & IoT:** Reinforcement Learning and Control (IE691), Convex Optimization (AAE561), Industrial IoT Implementation (ME597)
- Applied Mathematics & Data Science:** Artificial Intelligence (ECE570), Statistical Methods (STAT511), Applied Machine Learning, Numerical Computations
- Thermal & Energy Systems:** Distributed Energy Resources (ME597), Analysis of Thermal Systems (ME518), Advanced Thermodynamics (ME500), Refrigeration Systems